CIVITAS

Sustainable and smart mobility for all

2030

CIVITAS indicators

CO_2 emissions per km (ENV_FR_CO1)

DOMAIN

Transport	**Environment**	Energy	Society	Economy

TOPIC	**Freight**	
IMPACT	**Zero emission deliveries** *Improved environmental performance of urban freight distribution through a reduction in locally emitted CO_2 per kilometre travelled by delivery vehicles.*	**ENV_FR**

Category

Key indicator	Supplementary indicator	State indicator

CONTEXT AND RELEVANCE

Urban freight measures frequently aim to reduce the environmental impact of delivery operations while maintaining the ability of logistics operators to serve urban customers. A reduction in CO_2 emissions per kilometre indicates a cleaner vehicle mix, a shift towards locally zero-emission vehicles, improved fleet composition, or the use of urban freight solutions that reduce reliance on higher-emitting vehicle types.

This indicator measures the average tank-to-wheel CO_2 emissions associated with each kilometre travelled by delivery vehicles within a defined geographical area and reference period, using vehicle-kilometres travelled by vehicle type and standard theoretical emission factors. **It is a relevant indicator when the policy action aims to reduce the carbon intensity of urban freight vehicle movements. A successful action is reflected in a LOWER value of the indicator after the experiment compared to the BAU case.**

DESCRIPTION

The indicator assesses the average CO_2 emissions intensity of freight delivery vehicle activity. It reflects whether the kilometres travelled by delivery vehicles are increasingly performed by vehicle types with lower or zero local CO_2 emissions. The indicator is calculated by weighting the theoretical emission factor of each delivery vehicle type by the kilometres travelled by that vehicle type over the observation period, obtained by surveying delivery companies.

The indicator is expressed as **$kgCO_2/km$**.

METHOD OF CALCULATION AND INPUTS

The indicator should be calculated **endogenously** in the supporting tool. The user fills in the kilometres travelled per delivery vehicle type and the tool computes the indicator according to the method.

Method 1	
Survey delivery companies on kilometres travelled by delivery vehicles over a sample period	Significance: **0.75**

INPUTS

The following information is needed to compute the indicator:

- The **total kilometres travelled** by delivery vehicles in the pilot area during the chosen observation period, disaggregated by vehicle type.

The typical tank-to-wheel emission factors for selected vehicle types used by the tool in the calculation are:

- petrol vans: **72.5 gCO_2/km**;
- diesel vans: **79 gCO_2/km**;
- electric vans, hydrogen vans, cargo bikes and other alternative modes are considered locally zero-emission.

Source: EMEP/EEA air pollutant emission inventory guidebook 2023. https://www.eea.europa.eu/en/analysis/publications/emep-eea-guidebook-2023. Part B. 1.A.3.b.i-iv Road transport 2025. Tier 3-15.

EQUATIONS

The equation that is applied by the tool to quantify the indicator is:

$$\boldsymbol{CO_2KM} = \frac{\sum_i KM_i \times EF_i}{\sum_i KM_i}$$

Where:

$\boldsymbol{CO_2KM}$ = Average CO_2 emissions per kilometre travelled by delivery vehicles (kgCO_2/km)

KM_i = Total kilometres travelled by delivery vehicles of vehicle type i during the observation period (km)

EF_i = Theoretical tank-to-wheel CO_2 emission factor for vehicle type i (kgCO_2/km)

ALTERNATIVE INDICATORS

This indicator measures the CO_2 emissions intensity of kilometres travelled by urban freight delivery vehicles. To assess CO_2 emissions in relation to delivery activity, the CIVITAS Evaluation Framework also provides **ENV_FR_CO2**: CO_2 emissions per delivery, expressed in kgCO_2/delivery.